

INF2005 ACW1 Project Assignment Specification

Image and Audio Steganography, Digital Signatures and Security Verification

Demo dates: Week 5 (in your lab classes: **Tues** and **Thurs**, each team's demo schedule will be provided later). **Plan for a team demo not more than 25 mins, with all members involved.** Upload **your demo plan** (sequence of what to show, who is showing) to xSite with **all members duly signed Declaration of Originality form one day before the demo evaluation.**

1. Project Title

Steganographic Image and Audio Integrity Verification with Digital Signature-Based Authentication

2. Project Overview

This undergraduate project requires **student teams** to design, implement and demonstrate a **GUI-based LSB Replacement steganography program** (window-based or web-based) that protects and verifies both image and audio cover objects using steganography, hashing and digital signatures.

The project focuses on practical cybersecurity concepts: hiding a verification payload inside an image and an audio file, signing relevant verification data, extracting the hidden payload, checking the digital signature, and demonstrating positive and negative verification cases. *Video as a cover object is not required for the main assignment, but may be attempted as an optional challenge.*

3. Learning Outcomes

1. Explain how steganography can be used to embed hidden verification data in image and audio cover objects.
2. Apply hashing to generate integrity fingerprints for media files or verification payloads.
3. Use digital signatures to verify that a payload or file record was issued by a legitimate signer and has not been altered.
4. Implement encoding and decoding workflows for both image-based and audio-based hidden payloads.
5. Design positive and negative test cases for authentication, tamper detection and failed verification.
6. Design, explain and secure the start location (other than top-left corner) used to embed and extract the payload in each cover object.
7. Identify and justify innovation incorporated into the team's design or implementation.
8. Evaluate the security limitations of steganography, hashing and digital signatures.
9. Communicate technical work clearly through a structured demonstration with individual accountability.

4. Project Scenario

A small digital media verification team wants a lightweight tool that can protect an image file and an audio file before release, then later check whether each file is likely to be authentic. The tool should embed hidden verification information inside the cover object and use digital signature verification to confirm whether the embedded information is legitimate.

Students are required to build and demonstrate a prototype that can protect an image and an audio file, verify protected files, and detect selected tampering or failure conditions.

5. Mandatory Scope

- Required **cover objects: image and audio.** The image implementation may use PNG for the basic version. The audio implementation may use WAV/PCM for the basic version. The GUI able to play(execute) payload / display both cover and stego objects for comparison before and after encoding and decoding

- Required **security components**: steganographic payload, cryptographic hash and digital signature verification.
- Required **submission format**: with supporting source code (you cannot use online steganography portals), sample files, test evidence, README, declaration of originality form, and agreed team contribution/distribution statement.
- Required **25 mins demo duration**: All members must have allocated time to talk/demonstrate about his/her part/s. **Zero marks for absence**.
- Required cases: **at least two positive cases and at least three negative cases** across the image and audio workflows. At least one positive case and one negative case must be shown for each mandatory cover object, including:
 - cover object and payload **capacity check** (is payload size larger than cover object size?).
 - showing the **stego object being sent from party A to party B** (eg, via email), party B downloads to his/her folder and extracts the hidden message with proof of the message integrity, signature verification.
 - **various payload sizes** (message lengths) – use one of the **Learning Objectives** as short message, the Project Overview paragraph as large message and a relevant custom payload your team decide that can protect the hidden message’s confidentiality and integrity.
 - **selectable LSBs** from bits 1 to 8 of the cover object. Selection of number of LSBs to be implemented as part of GUI.
- Required **selectable payload start-location** design: the payload may be embedded beginning at any chosen start location in the image or audio cover object. The team must explain how the start location is designed or selected for each cover object, how the decoder identifies the same location for extraction, and how this information is secured against guessing, unauthorised extraction or tampering.
- Required **innovation explanation**: the team must explain what innovation was incorporated into its design or work, why it is *useful, meaningful and practical*, and how it improves the baseline implementation.
- Required **originality and contribution evidence**: teams must submit a signed declaration of originality and an agreed statement showing each member’s work contribution and contribution distribution.

6. Functional Requirements

Requirement	Description
FR1: Image input	The system must accept at least one standard image format, preferably PNG. JPEG may be used if the team explains compression-related limitations.
FR2: Audio input	The system must accept at least one standard audio format, preferably WAV/PCM for the basic implementation. Other formats may be used if the team explains encoding and compression limitations.
FR3: Payload generation	The system must create compact verification payloads containing media ID, timestamp, hash, nonce and team-defined metadata for image and audio cover objects.
FR4: Digital signature	The system must digitally sign the verification payload or an associated hash using a private key, and verify it using the corresponding public key.
FR5: Image steganographic embedding	The system must embed the payload and/or signature into the image using LSB replacement steganographic method.
FR6: Audio steganographic embedding	The system must embed the payload and/or signature into the audio cover object using LSB replacement steganographic method.
FR7: Variable start location	The payload may begin at any chosen start location in each cover object. The design must support identifying that location during extraction and must explain how the start-location information is protected.
FR8: Extraction and decoding	The system must extract the hidden payload and signature from both stego image and stego audio files.
FR9: Hash verification	The system must recompute the relevant media or payload hash and compare it with the embedded or signed value.
FR10: Verdict generation	The system must provide a clear result such as Authentic, Tampered, Signature Invalid, Payload Missing, Wrong Start Location, or Cannot Verify.
FR11: Positive and negative cases	The system must demonstrate successful verification and meaningful failure scenarios across both mandatory cover objects.
FR12: Evidence and reproducibility	The submitted files must allow the marker to understand and, where appropriate, reproduce the demonstrated workflow.
FR13: Innovation	The system must identify at least one design or implementation innovation beyond the simplest fixed-location LSB demonstration, or clearly justify why the chosen design is innovative for the team’s use case.

7. Required Security Workflow

Team may design their own implementation, but the following workflow is suggested for both image and audio cover objects:

1. User selects an original cover object: image or audio.
2. System computes a cryptographic hash of the cover object or selected stable representation.
3. System creates a payload containing media ID, timestamp, hash, nonce and team-defined metadata.
4. System digitally signs the payload or payload hash using a private key.
5. User selects or derives a payload start location* in the cover object, then embeds the payload and signature beginning at that location.
6. System outputs a stego image or stego audio file.
7. Verifier identifies* the correct start location, extracts the hidden payload and signature, and explains how that location was recovered or derived.
8. Verifier checks the signature using the public key.
9. Verifier recomputes the current media hash or relevant verification hash.
10. Verifier returns a verdict and explanation.
[*Show how the payload start location is selected or derived for image and audio, how the verifier finds it, and how the team secures this information.]

Suggested verdict categories: Authentic, Tampered, Signature Invalid, Payload Missing, Wrong Start Location, and Cannot Verify.

8. Optional Challenge

The following are optional extensions and should not be attempted at the expense of the mandatory image and audio requirements. Anyone of these below can be considered as the innovation component:

Optional Challenge	Description
Video cover object	Extend the system to support video as an additional cover object, such as selected-frame embedding or audio-track embedding.
Robust embedding	Improve payload survival under compression, noise, resampling or mild transformation using redundancy, error correction or spread-spectrum ideas.
Attack simulation module	Create a test module that simulates tampering, wrong-key verification, payload corruption, wrong start-location extraction or replay/substitution attempts.
Advanced start-location security	Derive the start location from a keyed pseudo-random function, encrypted header, seed phrase or other defensible method, and evaluate its limitations.
Steganalysis	Using known algorithms or methodologies, analyse a stego object and <i>convincing</i> infer the cover object is tampered or shows signs of hidden payload.

9. Required Submission Package

Item	Required Content
Deadline : one day before Demo day	
Demo plan	Planned demo for a duration of ≤ 25 mins (including members' 'air' time)
Declaration of originality	Signed declaration confirming that the work is original, sources are acknowledged, AI use is disclosed, and the submitted work reflects the team's own understanding.
Contribution/distribution statement	Agreed statement signed or acknowledged by all members showing each member's responsibilities and percentage contribution.
Deadline : week 5, Friday	
Source code	Complete implementation with clear folder structure.
README	Setup instructions, dependencies, commands and expected outputs.
Sample files	Original, protected and tampered image and audio files used in the demo.
Test evidence	Screenshots, logs or output files showing verification results.
Keys or key instructions	Public key and safe instructions for reproducing signature verification. Do not submit private keys that should remain secret in a real deployment unless generated only for the assignment demo.

10. Declaration of Originality and Contribution Distribution

Each team must submit a signed declaration of originality and an agreed contribution/distribution statement together with the project files. **Replace Px-x with your team number, eg P1-4.**

11. Assessment Rubric — 40 Marks

Team components: 35 marks. Individual components: 5 marks. Total: 40 marks.

	Criterion	Marks	What is assessed	Marking guide
1	Security design and problem framing / Team	5	Payload structure, start-location design, start-location security, and explanation of how the system verifies authenticity across both cover objects.	5 = clear and coherent; 3-4 = mostly clear; 1-2 = basic or weak; 0 = missing.
2	Image embedding, extraction and image case demonstration / Team	9	Working image encoder and decoder, including payload insertion, start-location recovery, extraction, positive image verification and negative/tampered image detection.	9-8 = fully working and clearly demonstrated; 5-7 = mostly working; 3-4 = partial or weak cases; 0-2 = not demonstrated or largely non-functional.
3	Audio embedding, extraction and audio case demonstration / Team	10	Working audio encoder and decoder, including payload insertion, start-location recovery, extraction, positive audio verification and negative/tampered audio detection.	10-9 = fully working and clearly demonstrated; 8-6 = mostly working; 3-5 = partial or weak cases; 0-2 = not demonstrated or largely non-functional.
4	Hashing, digital signature verification and failed-verification handling / Team	5	Appropriate use of hashing, payload or file signing, public-key verification, failed-signature or altered-payload handling, and clear linkage between verification logic and outcomes.	5 = correct, shown and explained; 3-4 = mostly correct; 1-2 = superficial or partly working; 0 = absent or incorrect.
5	Innovation incorporated / Team	4	Specific design or implementation improvement beyond the basic requirement, with usefulness and remaining limitations explained.	4 = implemented and justified; 3 = present with some value; 1-2 = claimed but weak; 0 = none.
6	Individual technical explanation / Individual	5	Each student explains their own technical contribution, and answers relevant demo questions with credible understanding.	5 = clear ownership and understanding; 3-4 = adequate; 1-2 = shallow or unclear; 0 = not credible or not shown.
7	Limitations, ethics and AI-use reflection / Team	2	Honest discussion of technical limits, responsible use, originality, and how AI was used and checked.	2 = clear and honest; 1 = brief but relevant; 0 = missing or misleading.

Use of Generative AI

SIT Internal



DO'S

- ✓ Use generative AI tools as a supplement, not a substitute, for your own learning and thinking. You can use them to explore different ideas, generate drafts, or check your work, but **NOT** to copy/ submit the generated content as your own
- ✓ Acknowledge and cite the generative AI tools you used in your assignment. You should also explain how you used the generated content and how you evaluated its quality and relevance.

DON'T'S

- ✗ Do not use generative AI tools to produce content that violates the academic integrity policy.
- ✗ Do not use generative AI tools to plagiarize or fabricate content from other sources
- ✗ Do not rely on generative AI tools to produce content that is accurate, relevant, and original
- ✗ It lacks critical thinking, creativity, and originality that are essential for academic work. Therefore, you should always verify, edit, and cite the generated content before using it in your assignment.